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Pharmacokinetics and Pharmacodynamics of Twice Daily and Once Daily Regimens of Empagliflozin in Healthy Subjects

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ABSTRACT

Purpose: This study was undertaken to compare the steady-state pharmacokinetic and pharmacodynamic properties of empagliflozin 5 mg twice daily (BID) and 10 mg once daily (QD) in healthy subjects.

Methods: In an open-label, 2-way crossover study, subjects (n = 16) received empagliflozin 5 mg BID for 5 days and empagliflozin 10 mg QD for 5 days in a randomized order, with a washout period of ≥6 days between each treatment. The primary objective was the comparison of the overall exposure during a 24-hour period at steady state ($AUC_{0-24,ss}$) for empagliflozin, based on standard bioequivalence criteria, with BID and QD dose regimens.

Findings: The study population comprised 7 (43.8%) men and 9 (56.3%) women with a baseline median age of 38.0 years (range, 23–47 years) and a median body mass index of 23.3 kg/m^2 (range, 19.8–27.8 kg/m^2). Based on standard bioequivalence criteria, there was no difference in the overall exposure of empagliflozin between BID and QD dose regimens (geometric mean ratio of $AUC_{0-24,ss}$ for empagliflozin 5 mg BID compared with empagliflozin 10 mg QD = 99.36%; 90% CI, 94.29–104.71). For empagliflozin 10 mg QD, mean (%CV) AUC during the dosing interval was 1900 nmol · h/L (20.6%), mean (%CV) $C_{max,ss}$ was 330 nmol/L (25.3%), and median (range) $T_{max,ss}$ was 1.0 hour (0.7–2.0 hours). For empagliflozin 5 mg BID, mean (%CV) AUC during the dosing interval was 1010 nmol · h/L (15.1%) and 867 nmol · h/L (18.6%) after the morning and evening dose, respectively, mean (%CV) $C_{max,ss}$ was 193 nmol/L (16.5%) and 120 nmol/L (21.0%), respectively, and median $T_{max,ss}$ was 1.0 hour (range, 0.7–2.0 hours) and 2.0 hours (range, 1.0–4.0 hours), respectively. The mean (%CV) cumulative amount of glucose excreted in urine during 24 hours was 52.1 g (32.1%) with empagliflozin 5 mg BID and 43.9 g (30.3%) with empagliflozin 10 mg QD. Adverse events were reported in six subjects (37.5%) receiving empagliflozin 5 mg BID and four (25.0%) receiving empagliflozin 10 mg QD. Headache was the most frequent AE. No severe, serious, or drug-related AEs were reported.

Implications: There were no clinically relevant differences in pharmacokinetic or pharmacodynamic properties between BID and QD dose regimens of empagliflozin in healthy subjects. Both dose regimens were well tolerated. EU Clinical Trials Register (EudraCT) number: 2009-012524-90. (*Clin Ther*. 2015;∎:∎∎∎–∎∎∎) © 2015 Elsevier HS Journals, Inc. All rights reserved.

Key words: empagliflozin, once daily, pharmacodynamics, pharmacokinetic properties, twice daily.

INTRODUCTION

Empagliflozin is a potent, selective inhibitor of sodium glucose cotransporter 2 (SGLT2).[1] By inhibiting SGLT2, empagliflozin reduces renal glucose reabsorption,[2] increasing urinary glucose excretion (UGE) and leading to a reduction in plasma glucose levels in patients with type 2 diabetes mellitus (T2DM).[3,4]

Metformin is a biguanide derivate that reduces hepatic glucose production via inhibition of gluconeogenesis.[5] It is recommended as the first-line pharmacologic treatment for patients with T2DM.[6] However, as diabetes progresses, metformin monotherapy often fails to maintain glycemic control.[6–8] Guidelines

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recommend the addition of a second anti-diabetes agent for patients not achieving glycemic targets with metformin monotherapy.[6,9] In addition, a combination of two oral anti-diabetes agents is a recommended initial treatment option for patients with high glycosylated hemoglobin (HbA1c) at diagnosis (eg, ≥9%), due to low probability of achieving target HbA1c levels with monotherapy.[6]

Compared with individual tablets, fixed-dose combinations of anti-diabetes agents reduce pill burden and have been shown to improve patient satisfaction and adherence,[10,11] and thus improve HbA1c.[10] Empagliflozin and metformin have complementary insulin-independent mechanisms of action and, in a Phase III trial, empagliflozin as add-on to metformin as individual tablets improved glucose control with a low risk of hypoglycemia and reduced body weight in patients with T2DM.[12] These indicate the potential for a fixed-dose combination of empagliflozin and metformin. As metformin immediate release is administered twice daily (BID), a fixed-dose combination of empagliflozin and metformin immediate release would require BID administration of empagliflozin. This study was undertaken to compare the pharmacokinetic and pharmacodynamic properties of multiple oral doses of empagliflozin given as a BID regimen versus a once daily (QD) regimen in healthy subjects.

METHODS

Participants

Eligible subjects were male and female, aged 18 to 50 years, with body mass index of 18.5 to 29.9 kg/m^2, and were generally in good health (as assessed by medical history, physical examination, vital signs, 12-lead electrocardiogram (ECG), and clinical laboratory tests).

Exclusion criteria included evidence of any clinically relevant concomitant disease, including central nervous system or psychiatric disorders; gastrointestinal surgery (except appendectomy); chronic or relevant acute infections; current smoker or alcohol or drug abuse; and participation in a trial with an investigational drug in the previous 2 months. All subjects gave signed informed consent before participating in the study.

Study Design

In an open-label, 2-way crossover study, subjects received empagliflozin 5 mg BID for 5 days and empagliflozin 10 mg QD for 5 days in a randomized order, with a washout period of ≥6 days between each treatment. Screening for eligibility took place 1 to 21 days before the first intake of study drug. Study drugs were given at the study site under the supervision of the investigator. Subjects visited the study site in the morning (and evening if they were receiving empagliflozin BID) of days 1 to 4 of each treatment period and were admitted to the study site on day 5 (pharmacokinetic and pharmacodynamic assessment day). Subjects were required to fast for ≥10 hours before study drug administration in the morning of day 1 and day 5. On day 5, study drugs were administered 2 hours before a light breakfast, and 30 minutes after dinner if receiving empagliflozin BID. Subjects were discharged on day 6 and returned to the study site for blood sampling that evening, and on the morning of days 7 and 8. All participants had an end-of-study examination 3 to 10 days after the last dose of study drug.

The study was conducted at the Human Pharmacology Centre, Boehringer Ingelheim Pharma GmbH & Co. KG, Biberach an der Riss, Germany, and was registered with the EU Clinical Trials Register (EudraCT number 2009-012524-90). The study was approved by the local Independent Ethics Committee (Baden-Württenberg, Stuttgart) and the German Competent Authority (the Federal Institute for Drugs and Medical Devices [BfArM]). It was carried out in compliance with the protocol and the principles of the Declaration of Helsinki, in accordance with the International Conference on Harmonization Harmonized Tripartite Guideline for Good Clinical Practice.

End Points

The primary end points were the relative overall exposure to empagliflozin BID versus QD based on the AUC over a uniform dosing interval (τ) at steady state ($AUC_{\tau,ss}$) for empagliflozin 10 mg QD, the AUC at steady state over two dosing intervals ($AUC_{0-24,ss}$) for empagliflozin 5 mg BID on day 5, and the diurnal variability in the rate and extent of exposure to empagliflozin 5 mg BID after the morning and evening doses, based on $AUC_{\tau,ss}$ and C_{max} at steady state ($C_{max,ss}$).

Secondary pharmacokinetic end points included time to $C_{max,ss}$ ($T_{max,ss}$) and $t_{½}$ at steady state ($t_{½,ss}$). Secondary pharmacodynamic end points were cumulative UGE over 12 hours (UGE_{0-12}) and

24 hours (UGE_{0-24}) for empagliflozin 5 mg BID, and cumulative UGE over 24 hours (UGE_{0-24}) for empagliflozin 10 mg QD. Safety profile and tolerability were evaluated based on adverse events (AEs), changes in vital signs (blood pressure and pulse rate), 12-lead ECG, clinical laboratory tests, physical examination, and a global assessment of tolerability by the investigator.

Pharmacokinetic Analysis

The pharmacokinetic profile of empagliflozin with QD regimen was determined using blood samples (2.7 mL) collected at predose and postdose at 0.33, 0.67, 1, 1.5, 2, 3, 4, 6, 8, 10, 12, 14, 24, 36, 48, and 72 hours on day 5. For empagliflozin BID, blood samples were collected at predose and postdose at 0.33, 0.67, 1, 1.5, 2, 3, 4, 6, 8, 10 (morning dose only) and 12 hours relative to both morning and evening doses. Additional blood samples were collected at 24, 36, and 60 hours postdose, relative to the evening dose on day 5 for empagliflozin BID. Morning predose samples were collected on days 1 to 4 from all subjects.

Empagliflozin concentrations in plasma were analyzed using a validated HPLC-MS/MS assay with a solid-phase−supported liquid extraction process. Calibration curves were created using weighted (1/×2) quadratic regression and results were calculated using peak area ratios. An acceptable standard curve was to contain ≥75% of the originally specified standard calibrator samples in the final regression set, including one or more samples each at the upper and lower limits of quantitation. Mean r^2 for calibration curves was 0.999. For all methods, analysis of quality-control samples demonstrated acceptable precision and accuracy (≥67% of samples were within 15.0% of their respective nominal values, and ≥50% of samples at each concentration met the same 15.0% criteria).

Noncompartmental pharmacokinetic parameters were calculated using WinNonlin™ software (version 5.2; Pharsight Corporation, Mountain View, California, USA). C_{max} and T_{max} values were determined directly from the plasma concentration-time profiles for each subject and $t_{½,ss}$ was calculated as the quotient of ln(2) and the apparent terminal rate constant. AUCs were calculated using the linear trapezoidal method for ascending concentrations and the log trapezoidal method for descending concentrations.

Pharmacodynamic Analysis

All urine voided during the sampling intervals on day 5 was collected. After empagliflozin 5 mg BID, sampling intervals were 0 to 4, 4 to 8, and 8 to 12 hours after each administration of study drug. After empagliflozin 10 mg QD, sampling intervals were 0 to 4, 4 to 8, 8 to 12 and 12 to 24 hours after each administration of study drug. Glucose concentrations were analyzed by the Glucose Hexokinase enzymatic method. Cumulative UGE (mg) was calculated from the UGE values for each relevant sampling period, which were calculated as glucose concentration (mg/dL) × urine volume (dL).

Safety Analysis

Subjects were monitored for AEs throughout the study; AEs were considered on treatment if they occurred after the first dose of study drug and up to 60 hours after the last dose, and were coded using the Medical Dictionary for Drug Regulatory Activities (version 12.1). A 12-lead ECG was performed at screening and at the end-of-study examination. Vital signs and clinical laboratory tests were conducted at screening, before first drug administration on day 1, and at the end-of-study examination.

Statistical Analysis

All subjects who provided one or more observations for one or more primary pharmacokinetic end point without any protocol violations relevant to the evaluation of relative overall exposure were included in the analysis of primary end points (pharmacokinetic set). Relative overall exposure to empagliflozin BID and QD was analyzed using an analysis of variance (ANOVA) model on the logarithmic scale with treatment, period, and sequence as fixed effects, and subject within sequence as a random effect. The difference in least-square means and respective confidence intervals (CIs) for $AUC_{0-24,ss}$ were back-transformed to the original scale to obtain the geometric mean ratio and corresponding 2-sided 90% CIs. Standard bioequivalence criteria were applied (80%−125% for the 90% CI of the geometric mean ratio). No formal hypothesis testing was performed. Diurnal variability in $AUC_{\tau,ss}$ and $C_{max,ss}$ for empagliflozin 5 mg BID after the morning and evening doses was analyzed descriptively. Statistical analyses were performed using SAS software, version 8.2 (SAS Institute, Cary, North Carolina, USA). UGE was assessed descriptively. Safety profile was assessed descriptively in the treated set comprising all participants who took one or more doses of study drug.

RESULTS

Participants

Sixteen subjects participated in the study (n = 7 [43.8%] men and n = 9 [56.3%] women). Median age was 38.0 years (range, 23–47 years) and median body mass index was 23.3 kg/m^2 (range, 19.8–27.8 kg/m^2). All subjects completed the study and were included in the pharmacokinetic, pharmacodynamic, and safety analyses.

Pharmacokinetics

Steady-state plasma concentrations of empagliflozin were achieved by day 5 after BID and QD administration of empagliflozin. Empagliflozin 5 mg BID and 10 mg QD were rapidly absorbed, reaching peak levels between 1.0 and 2.0 hours post dose (Figure 1 and Table I). Thereafter, plasma levels declined in a biphasic fashion, with a rapid distribution phase and a slower elimination phase (Figure 1).

The pharmacokinetic parameters of empagliflozin 5 mg BID and 10 mg QD are shown in Table I. The exposure to empagliflozin 5 mg BID ($C_{max,ss}$ and $AUC_{\tau,ss}$) was slightly lower after the evening dose compared with the morning dose, and peak levels were achieved sooner after the morning dose (Table I). There was no significant difference in overall exposure (AUC) to empagliflozin between empagliflozin 5 mg BID and empagliflozin 10 mg QD regimens according to standard bioequivalence criteria (geometric mean ratio = 99.36% [90% CI, 94.29–104.71]). Intraindividual variability (%CV) for $AUC_{0-24,ss}$ between the empagliflozin BID and QD regimens was 8.4%.

Pharmacodynamics

Mean cumulative UGE_{0-24} after empagliflozin BID and QD regimens are shown in Figure 2. Mean (%CV) UGE_{0-24} was 52.1 g (32.1%) after empagliflozin 5 mg BID and 43.9 g (30.3%) after empagliflozin 10 mg QD. With the BID regimen, mean (%CV) UGE_{0-12} was 28.8 g (36.0%) after the morning dose and 23.3 g (30.5%) after the evening dose.

Safety and Tolerability

AEs were reported in 6 subjects (37.5%) when receiving empagliflozin 5 mg BID and four subjects (25.0%) when receiving empagliflozin 10 mg QD. No severe, serious, or investigator-defined drug-related AEs were reported. Only 2 subjects (12.5%) reported AEs of moderate intensity (headache). Headache was the most frequent AE, reported by 5 subjects (31.3%) when receiving empagliflozin 5 mg BID and 1 subject (6.3%) when receiving empagliflozin 10 mg QD. All AEs had resolved by the end of the study. Polyuria or nocturia was not reported as an AE. There were no clinically relevant changes in clinical laboratory findings, vital signs, or ECG. Global tolerability was assessed as “good” for all subjects in both treatment sequences by the investigator.

DISCUSSION

Metformin is recommended as the first-line pharmacologic treatment for patients with T2DM, but the addition of a second anti-diabetes agent is often required to maintain glycemic control.[6] Fixed-dose combinations of anti-diabetes agents can improve patient adherence compared with individual tablets.[10] The complementary mechanisms of action of empagliflozin and metformin and the improvements in glycemic control when empagliflozin was given as add-on to metformin[12] indicate the potential of a fixed-dose combination of metformin and empagliflozin. As a fixed-dose, combination of metformin immediate release and empagliflozin would require BID administration of empagliflozin; this study was undertaken to compare the pharmacokinetic and pharmacodynamic properties of empagliflozin 5 mg BID and 10 mg QD at steady state in healthy subjects. Overall exposure over 24 hours was similar between the BID and QD empagliflozin regimens, with low intra-individual variability. Although the rate and extent of exposure to the empagliflozin 5 mg BID regimen was slightly lower with the evening dose than the morning dose, they were similar to those for the empagliflozin 10 mg QD regimen during a 24-hour period. Indeed, there were no clinically relevant differences in the pharmacokinetic properties of empagliflozin 5 mg BID and 10 mg QD. The pharmacokinetic properties of empagliflozin in this study were consistent with the steady-state pharmacokinetic properties of empagliflozin 10 mg QD in patients with T2DM.[3,4]

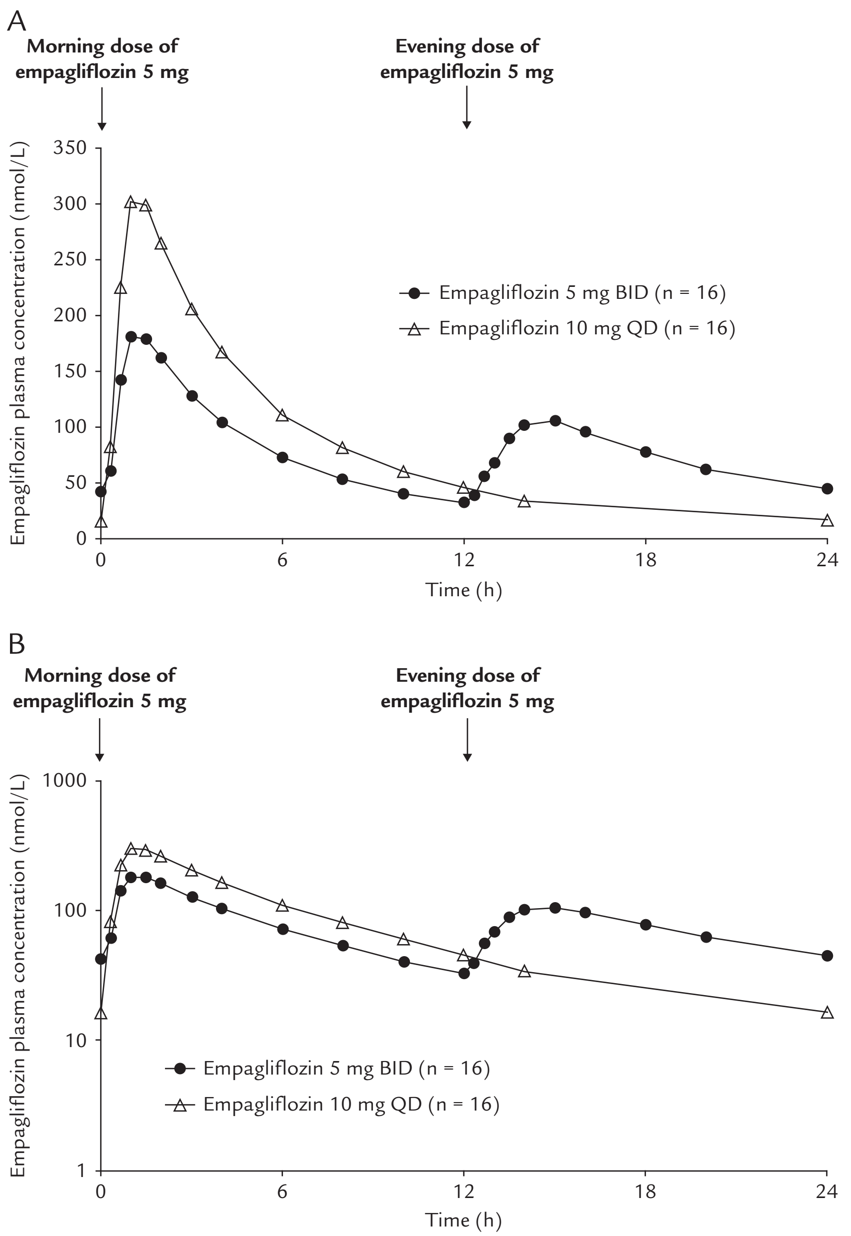

Figure 1. Mean plasma concentration-time profiles of empagliflozin 5 mg BID and empagliflozin 10 mg QD at steady state. (A) Linear scale. (B) Semi-log scale.

Consistent with its mode of action and with other data in healthy subjects,[13] empagliflozin increased UGE. UGE_{0-24} appeared to be slightly higher with empagliflozin 5 mg BID than with empagliflozin 10 mg QD, and slightly lower with the evening dose than the morning dose for empagliflozin 5 mg BID. These small differences in UGE could be due to the small differences in drug exposure between the regimens and/or administration of the morning dose 2 hours before breakfast and the evening dose 30 minutes after dinner. Although administration with food has no clinically relevant effects on the pharmacokinetic properties of empagliflozin, a study has shown that a high-fat, high-calorie meal slightly

Table I. Pharmacokinetic properties of empagliflozin 5 mg BID and 10 mg QD (n = 16). Data are given as mean (%CV) unless otherwise specified.

Parameter	Empagliflozin 5 mg BID		Empagliflozin 10 mg QD
	Morning Dose	Evening Dose	
$AUC_{\tau,ss}$, nmol · h/L*	1010 (15.1)	867 (18.6)	1900 (20.6)
$C_{max,ss}$, nmol/L	193 (16.5)	120 (21.0)	330 (25.3)
$T_{max,ss}$, h, median (range)	1.0 (0.7–2.0)	2.0 (1.0–4.0)	1.0 (0.7–2.0)
$t_{½,ss}$, h†	–	14.0 (55.4)	13.6 (43.0)

ss = steady state.
*5 mg BID: $AUC_{0-12,ss}$; 10 mg QD: $AUC_{0-24,ss}$.
†Duration of sampling was not sufficient to estimate $t_{½}$ after the morning dose.

reduced overall exposure and reduced C_{max} by 37% compared with fasted conditions,[14] similar to the differences between the evening and morning doses in this study. Overall, however, there was no clinically relevant difference in UGE_{0-24} associated with empagliflozin 5 mg BID and 10 mg QD in this study. Therefore, the slight effect of food on empagliflozin overall exposure and C_{max} is not considered to be clinically relevant for a potential fixed-dose combination of empagliflozin and metformin. A potential limitation of this study is the administration of the morning dose of empagliflozin 2 hours before breakfast, because metformin must be administered with meals to limit gastrointestinal side effects. Of note, in patients with T2DM on a background of metformin, administration of empagliflozin 5 mg BID and empagliflozin 12.5 mg BID for 16 weeks resulted in comparable and statistically

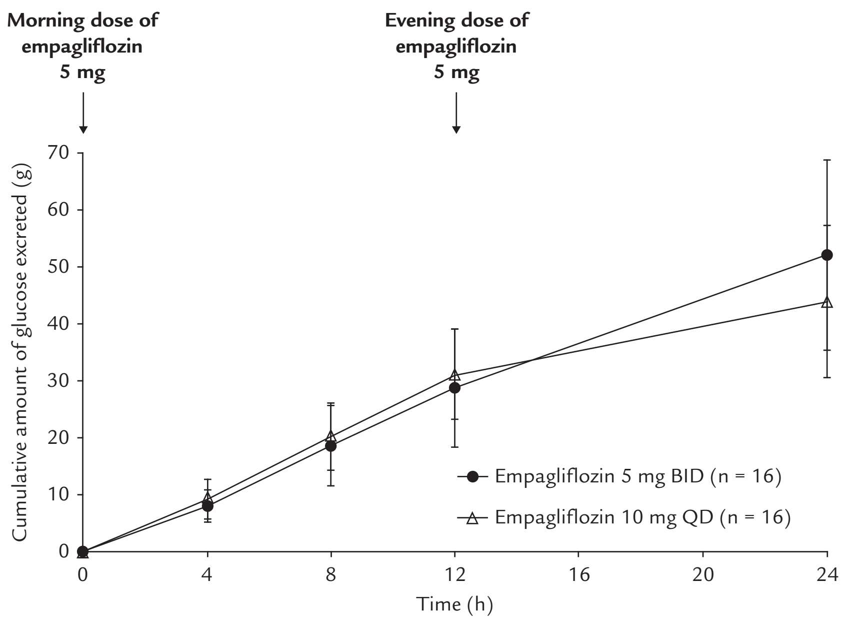

Figure 2. Mean (SD) cumulative urine glucose excretion over 24 hours (UGE_{0-24}) after empagliflozin 5 mg BID and empagliflozin 10 mg QD.

noninferior HbA1c reductions compared with empagliflozin 10 mg QD and empagliflozin 25 mg QD, respectively.[15]

CONCLUSIONS

Overall exposure to empagliflozin over 24 hours was similar between empagliflozin 5 mg BID and 10 mg QD dosing regimens. There were no clinically relevant differences in pharmacokinetic or pharmacodynamic properties between empagliflozin 5 mg BID and 10 mg QD in healthy subjects and both regimens were well tolerated.

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Sreeraj Macha and Tobias Brand made substantial contributions to the study conception and design and were involved in the acquisition of data; all authors were involved in the analysis and interpretation of data. The authors were fully responsible for all content and editorial decisions, were involved at all stages of manuscript development, and have approved the final version.

CONFLICTS OF INTEREST

Sreeraj Macha was an employee of Boehringer Ingelheim at the time the study was conducted. Tobias Brand, Thomas Meinicke, Jasmin Link, and Uli C. Broedl are employees of Boehringer Ingelheim. The authors have indicated that they have no other conflicts of interest regarding the content of this article.

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